A LOCALLY CONSTANT SHEAF OF MODULES ON A CONNECTED SCHEME IS CONSTANT AFTER AN ETALE COVERING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument. Its only outside input is the definition of locally constant, namely triviality on the members of an etale covering; the author states this at the start of the proof and then carries out the argument.

Editorial note. The label lemma-connected-locally-constant occurs in no other staged Stacks chapter, so no cross-chapter restatement had to be weighed for this card.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Locally constant sheaf of Lambda-modules on the small etale site. Same chapter, the definition of a locally constant sheaf of Lambda-modules on X_etale (constant on the members of an etale covering). Supplies the covering {U_i -> X} on whose members the sheaf is constant, which is the starting datum of the counted proof.
- Constant sheaf underline M associated to a Lambda-module and its restriction to etale open subschemes. Same chapter, the constant sheaf underline M on X_etale and the identification of its restriction to U_i with underline M_i. Supplies the comparison sheaves underline M_i and the target sheaf underline M of the conclusion.
- Etale morphisms are open. Same chapter (also used throughout the chapter), the openness of etale morphisms. Makes U_M = union of the images Im(U_i -> X) an open subset of X, so that X = coproduct U_M is an open covering.

Lemma 0.1. *Let X be a connected scheme. Let Λ be a ring and let $\mathcal{F}$ be a locally constant sheaf of Λ-modules. Then there exists a Λ-module M and an étale covering $\{U_i \to X\}$ such that $\mathcal{F}|_{U_i} \cong \underline{M}|_{U_i}$.*

Proof. Choose an étale covering $\{U_i \to X\}$ such that $\mathcal{F}|_{U_i}$ is constant, say $\mathcal{F}|_{U_i} \cong \underline{M_i}_{U_i}$. Observe that $U_i \times_X U_j$ is empty if M_i is not isomorphic to M_j. For each Λ-module M let $I_M = \{i \in I \mid M_i \cong M\}$. As étale morphisms are open we see that $U_M = \bigcup_{i \in I_M} \mathrm{Im}(U_i \to X)$ is an open subset of X. Then $X = \coprod U_M$ is a disjoint open covering of X. As X is connected only one U_M is nonempty and the lemma follows. □

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.